ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Minerals and Energy Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** manufacturing of steel and ferro-manganese alloy	Alloy
2	**(a)** aluminium	Non-ferrous
3	**(c)** Madhya Pradesh	Balaghat
4	**(d)** falling water	Dams
5	**(d)** Chhattisgarh	Iron ore
6	**(c)** Odisha	Panchpatmali
7	**(a)** cement industry	Also in smelting iron
8	**(b)** nuclear power station	Maharashtra
9	**(a)** about three-fourths	Iron, manganese, nickel
10	**(a)** Karnataka	Western Ghats

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Solar energy is energy from the sun converted into electricity by photovoltaic technology (1); It is a renewable and pollution-free source (1).	1+1
12	Mumbai High is an offshore oilfield in the Arabian Sea off Mumbai (1); It is the largest petroleum-producing region of India (1).	1+1
13	Natural gas is a clean fossil fuel found with petroleum deposits (1); It is used as a fuel, as CNG and as an industrial raw material (1).	1+1
14	Minerals are finite and non-renewable and take millions of years to form (1); Conservation ensures their availability for the future (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Minerals occur in veins and lodes in igneous and metamorphic rocks (1); In sedimentary rocks they occur in beds or layers (1); Some form by decomposition of rocks, leaving residual weathered material, e.g., bauxite (1).	1+1+1
16	Peat is the first stage with low carbon and high moisture (1); Lignite is low-grade brown coal and bituminous is the most popular commercial coal (1); Anthracite is the highest quality hard coal (1).	1+1+1
17	Haematite has 60 to 70 per cent iron and is the most important ore (1); Magnetite has over 70 per cent iron and is the best quality ore (1); Magnetite has magnetic qualities and is used in the electrical industry (1).	1+1+1
	OR: Bauxite is the ore from which aluminium is extracted (1); Aluminium is light, strong and a good conductor, so it is used in aircraft and electricals (1); Odisha leads in bauxite production, Panchpatmali being a major deposit (1).	1+1+1
18	Minerals are finite and non-renewable (1); Rich deposits are few, and costly to extract when low in grade (1); Recycling and substitutes help to conserve them (1).	1+1+1
	OR: The Odisha-Jharkhand belt has Mayurbhanj and Kendujhar (1); The Durg-Bastar-Chandrapur belt includes Bailadila in Chhattisgarh (1); The Ballari-Chitradurga-Chikkamagaluru-Tumakuru belt includes Kudremukh in Karnataka (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Mica is a non-metallic mineral used in electrical and electronic industries as insulator (1); It occurs in Jharkhand's Koderma-Gaya-Hazaribagh belt, Andhra and Rajasthan (1); Limestone is found in calcium carbonate rocks (1); It is the raw material for cement and smelting iron (1); It is found in sedimentary rocks, mainly in Gujarat, Rajasthan and Madhya Pradesh (1).	1+1+1+1+1

	OR: Coal is the most abundant fossil fuel in India (1); Types are peat, lignite, bituminous and anthracite (1); Gondwana coal is found in the Damodar, Mahanadi and Godavari valleys (1); Lignite is found in Neyveli (1); Coal is used for power and industry (1).	1+1+1+1+1
20	Bauxite is the ore of aluminium (1); Aluminium is light, strong and a good conductor (1); Odisha is the leading producer of bauxite (1); Copper is malleable and ductile and used in electrical goods (1); Madhya Pradesh is the largest copper producer (1).	1+1+1+1+1
	OR: Use public transport (1); Switch off unused lights (1); Use power-saving devices (1); Check electrical equipment regularly (1); Use non-conventional sources (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that **hematite** has more iron than magnetite; **magnetite** is richer (up to 70 per cent), though hematite is used more in quantity.
- Mixing the order of coal: peat, lignite, bituminous, **anthracite** (highest carbon, best quality).
- Classifying **hydel power** as non-conventional; hydel power, coal, petroleum, firewood and natural gas are conventional.
- Calling minerals renewable; they are exhaustible and take millions of years to form.
- Confusing **veins** and **lodes** (igneous and metamorphic rocks) with **placer deposits** (alluvial sands) and sedimentary beds.
- Forgetting the sources of nuclear minerals: **uranium and thorium** occur in Jharkhand and the Aravalli ranges of Rajasthan, and thorium is also found in the **monazite sands of Kerala**.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Minerals and Energy Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** malleable, ductile and a good conductor	Electrical industry
2	**(a)** photovoltaic technology	Panels
3	**(c)** 60-70% iron	Most important
4	**(c)** manganese	India
5	**(b)** Neyveli	Tamil Nadu
6	**(b)** anthracite	80% carbon
7	**(c)** Tamil Nadu	Nagercoil to Madurai
8	**(b)** Digboi	Assam
9	**(c)** atomic power	Monazite sands in Kerala
10	**(d)** homogeneous naturally occurring substance with definable internal structure	Definition

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Ferrous minerals contain iron, such as iron ore and manganese (1); Non-ferrous minerals do not contain iron, such as copper, bauxite and gold (1).	1+1
12	Hydroelectric power is electricity generated by the force of falling water turning turbines (1); Bhakra-Nangal and the Damodar Valley projects are examples (1).	1+1
13	A mineral is a homogeneous, naturally occurring substance with a definable internal structure (1); Minerals are found in varied forms from hard diamond to soft talc (1).	1+1
14	Nuclear energy is obtained by altering the structure of atoms of uranium and thorium (1); It is used to produce electricity, e.g., Tarapur (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Minerals are metallic and non-metallic (1); Metallic minerals are ferrous (iron ore, manganese) and non-ferrous (copper, bauxite) (1); Non-metallic minerals include mica, limestone and also energy minerals such as coal and petroleum (1).	1+1+1
16	Manganese is used in the manufacture of steel and ferro-manganese alloy (1); It is also used in bleaching powder, insecticides and paints (1); Odisha is the leading producer (1).	1+1+1
17	Biogas is produced from shrubs, farm waste, animal and human waste (1); It is used for cooking and lighting in rural areas (1); The residue is a good manure (1).	1+1+1
	OR: Natural gas is a clean energy resource found with petroleum (1); It is used as an industrial fuel and as a source of power (1); It is used as CNG in vehicles and as a chemical feedstock (1).	1+1+1
18	Petroleum provides fuel for heat and lighting and lubricants for machinery (1); It is a raw material for many manufacturing industries (1); Petrochemicals produce fertilisers, synthetic fibres, plastics and pesticides (1).	1+1+1
	OR: Mica is a non-metallic mineral used in electrical and electronic industries (1); It is a good insulator and withstands high voltage (1); Deposits occur in Jharkhand, Andhra Pradesh and Rajasthan (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Hydroelectricity uses falling water to turn turbines (1); It is renewable and pollution-free (1); Thermal power uses coal, petroleum and natural gas (1); Thermal plants are located near coalfields (1); Both are conventional sources (1).	1+1+1+1+1
	OR: Minerals are the raw materials of industry (1); Iron ore supports the steel industry (1); Coal and oil provide energy (1); Limestone feeds the cement industry (1); Without them industrial growth would stop (1).	1+1+1+1+1

20	Iron ore occurs as haematite and magnetite, magnetite being the best (1); Belts are Odisha-Jharkhand, Durg-Bastar-Chandrapur and Ballari-Chitradurga (1); Bailadila, Mayurbhanj and Kudremukh are important (1); Manganese is used in steel and Odisha is the leading producer (1); Iron ore is exported to Japan and South Korea (1).	1+1+1+1+1
	OR: Petroleum is found in anticlines and fault traps (1); Mumbai High, Gujarat and Assam are producers (1); Digboi is the oldest oilfield (1); Natural gas is found with oil and in Krishna-Godavari basin (1); It is used as CNG and for fertilisers (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that **hematite** has more iron than magnetite; **magnetite** is richer (up to 70 per cent), though hematite is used more in quantity.
- Mixing the order of coal: peat, lignite, bituminous, **anthracite** (highest carbon, best quality).
- Classifying **hydel power** as non-conventional; hydel power, coal, petroleum, firewood and natural gas are conventional.
- Calling minerals renewable; they are exhaustible and take millions of years to form.
- Confusing **veins** and **lodes** (igneous and metamorphic rocks) with **placer deposits** (alluvial sands) and sedimentary beds.
- Forgetting the sources of nuclear minerals: **uranium and thorium** occur in Jharkhand and the Aravalli ranges of Rajasthan, and thorium is also found in the **monazite sands of Kerala**.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Minerals and Energy Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** non-metallic mineral	Insulator
2	**(a)** bauxite	Residual mass
3	**(b)** ferrous mineral	Contains iron
4	**(b)** clean fossil fuel	CNG
5	**(c)** mineral oil	Crude
6	**(c)** most abundantly available fossil fuel in India	Power and industry
7	**(b)** Meghalaya	Narrow tunnels
8	**(c)** calcium carbonate	Cement raw material
9	**(c)** magnetite	70% iron content
10	**(a)** 200 million years old	Damodar valley

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	An ore is an accumulation of any mineral mixed with other elements (1); Metals are extracted from ores after mining and refining (1).	1+1
12	Rat-hole mining is a method of extracting coal in Meghalaya through narrow horizontal tunnels (1); It is unsafe because tunnels can collapse or flood (1).	1+1
13	Haematite is the most important industrial iron ore, with 60 to 70 per cent iron (1); It is used in the iron and steel industry (1).	1+1
14	The ranks of coal are peat, lignite, bituminous and anthracite (1); Anthracite is the highest quality coal (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hydroelectricity is generated by fast flowing water turning turbines (1); It is renewable because the water cycle renews water (1); Examples are Bhakra Nangal and the Damodar Valley Corporation (1).	1+1+1
16	Limestone is found in association with rocks composed of calcium carbonate (1); It is the basic raw material for the cement industry (1); It is also essential for smelting iron ore in blast furnaces (1).	1+1+1
17	Solar energy is converted into electricity by photovoltaic cells (1); India is a tropical country with abundant sunshine (1); Wind energy farms are widespread, especially in Tamil Nadu from Nagercoil to Madurai (1).	1+1+1
	OR: Mining brings out minerals but is dangerous for workers (1); Miners suffer lung diseases and mine collapses or floods (1); Mining also pollutes land, water and air (1).	1+1+1
18	Coal is the most abundantly available fossil fuel in India (1); It is used for power generation and in industry (1); Gondwana coal, about 200 million years old, is found in the Damodar valley (1).	1+1+1
	OR: Copper is malleable, ductile and a good conductor of heat and electricity (1); It is used in electrical cables, electronics and chemical industries (1); Madhya Pradesh's Balaghat mines are the leading producers (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Minerals are exhaustible and take millions of years to form (1); Rich deposits are limited (1); Mining and refining are costly (1); Judicious use, recycling and substitutes are needed (1); Conservation ensures supply for future generations (1).	1+1+1+1+1
	OR: Nuclear power uses uranium and thorium (1); Tarapur, Kota and Kalpakkam are stations (1); Solar energy uses photovoltaic cells (1); Wind energy uses windmills in Tamil Nadu (1); Biogas and tidal energy are also non-conventional (1).	1+1+1+1+1

20	Minerals occur in veins and lodes in igneous and metamorphic rocks (1); In sedimentary rocks they occur in beds or layers (1); Minerals may be metallic or non-metallic (1); Metallic minerals are divided into ferrous and non-ferrous (1); Energy minerals include coal and petroleum (1).	1+1+1+1+1
	OR: Conventional sources are coal, petroleum, natural gas and hydel power (1); Coal is the main source of thermal power (1); Petroleum and gas are used in transport and industry (1); Hydel power is renewable (1); Nuclear power adds to supply (1).	1+1+1+1+1

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